Assignment 2:
Predicting GitHub User Activity

Ella Tso · Student ID 113700058

Code: github.com/ellatso/NYCU-BigData-Assignment2

Big Data Analytics Techniques
and Applications

Fall 2026

Overview

LightGBM on time-aware relational features built from rows visible at prediction time

- **Task** for each of 4,465 test users, predict whether they open_pr, open_issue, issue_comment, pr_review, pr_review_comment, push during 11/22–11/29 (Taiwan).
- **Data** GH Archive hourly files → official preprocess.py → 8 relational tables (9,000-user cohort, anonymized).
- **Method** weekly as-of samples → temporal + relational (FK-join) features → 6 LightGBM heads → per-activity thresholds.
- **Evaluation** Score = (macro-F1 + micro-F1) / 2 on later weeks; train / calibration / development split by time; 11 ablations vs. copy-last-week.
- **Code** github.com/ellatso/NYCU-BigData-Assignment2

1,608
hourly GH Archive files 09/01–11/07 UTC

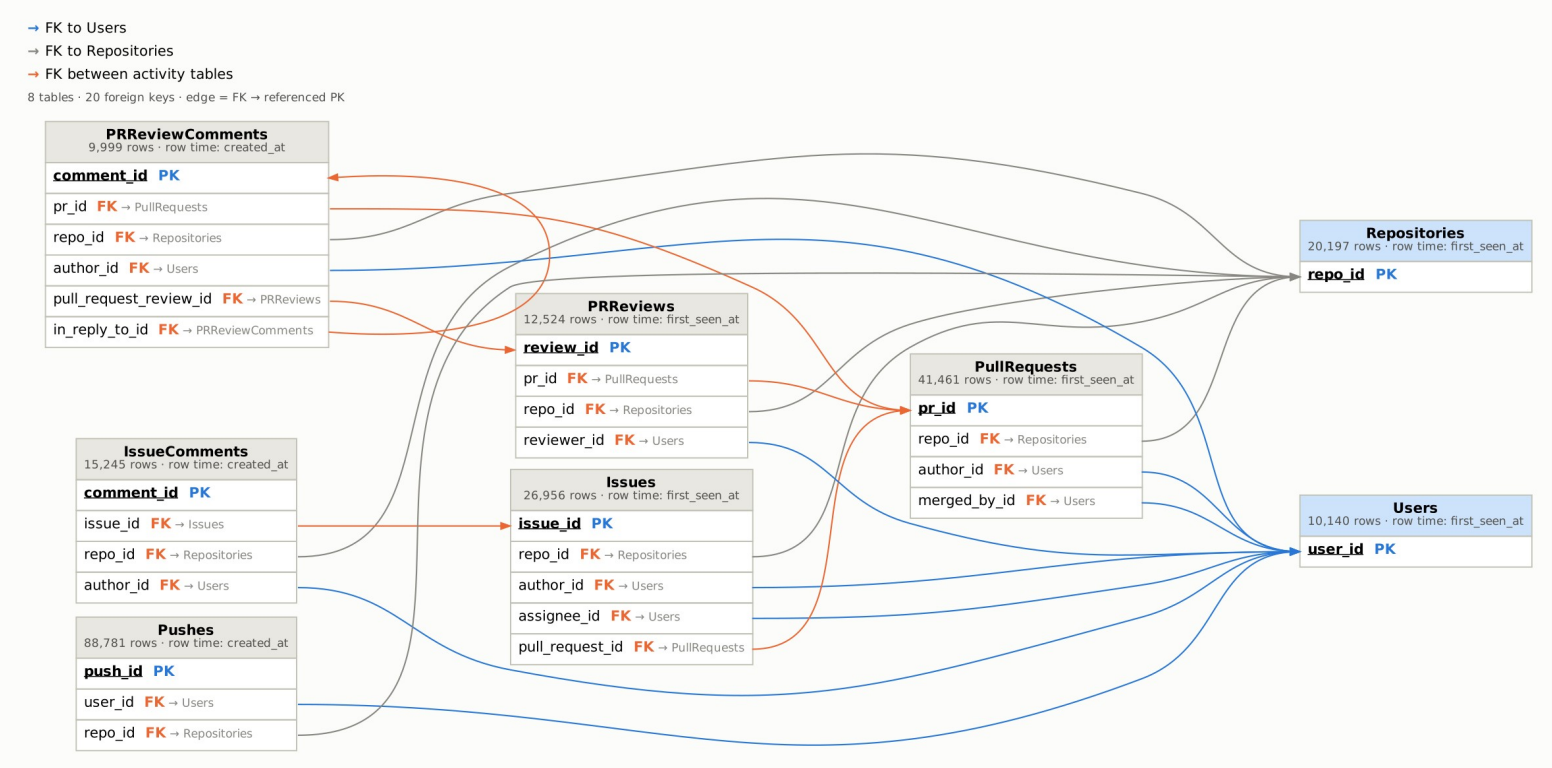
11
matched ablations with bootstrap 95% CIs

+1.6 pt
best temporal variants vs. copy-last-week (CI excludes 0)

Identical
CSV after an independent rebuild + refit

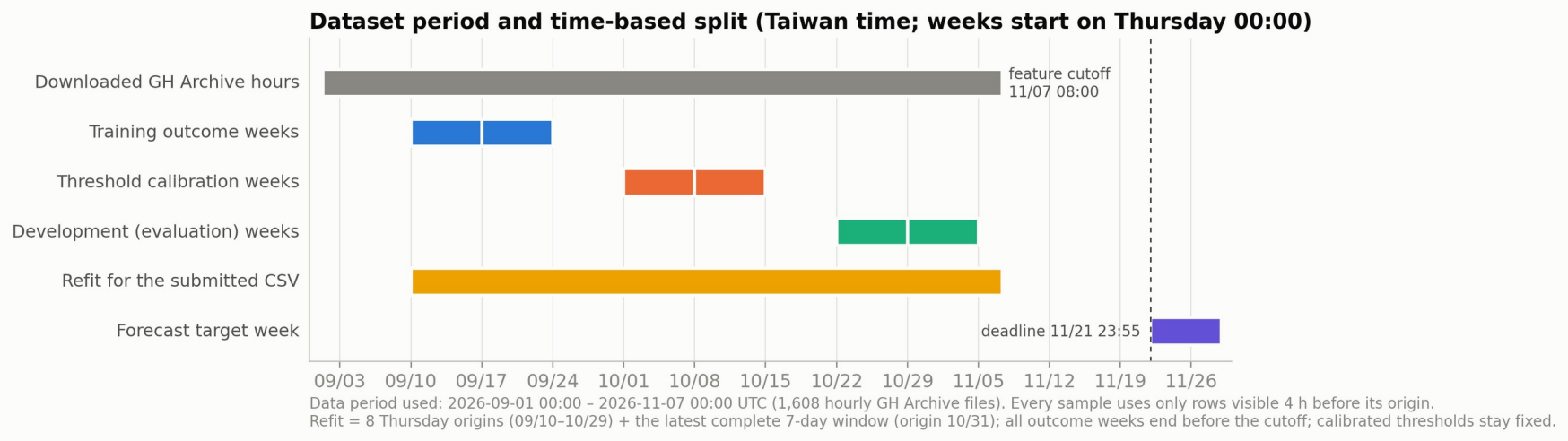
(1%) Question 1 Schema graph

8 tables as nodes and 20 foreign keys as edges (FK → referenced PK)



(1%) Question 1 Dataset time period

Data period used: 2026-09-01 00:00 – 2026-11-07 00:00 UTC (1,608 hourly GH Archive files)



1,608 h
all hourly files downloaded & gzip-verified (28.7 GB)

8 tables
official preprocess.py; PK/FK checks pass

9 origins
final refit 09/10–10/31 (Taiwan, outcome ≤ cutoff)

352 h
lead from the feature cutoff to the target week

(3%) Question 2 Relational features from FK joins

Time-windowed FK aggregations turn the 8-table graph into user-level signals

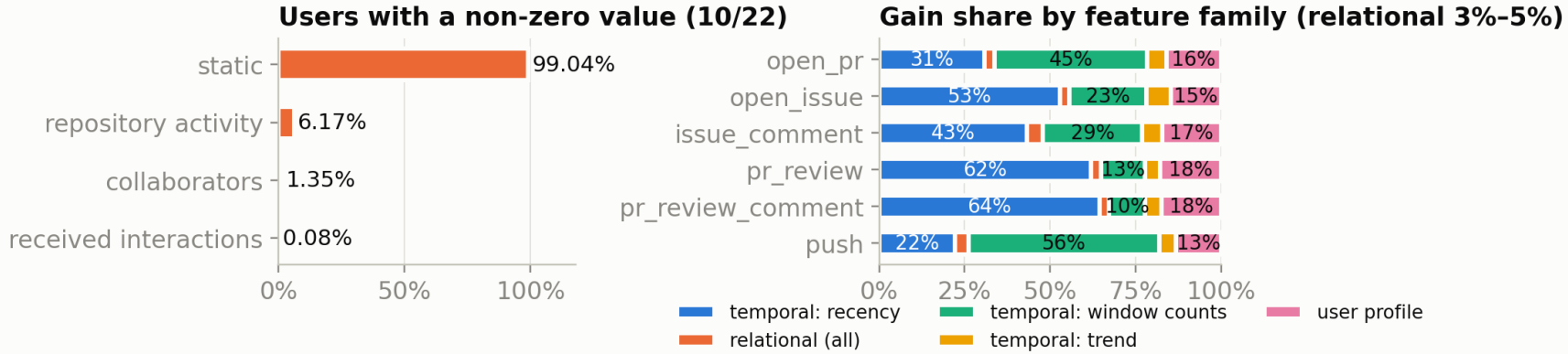
Family	FK path	What is computed	#
Repository activity	user → events → Repositories ← events ← other users	Others' activity per type in repos the user touched (90 d); 7-day mean / max; own events removed	12
Reviewed-repo PRs	user → PRReviews → Repositories ← PullRequests	PRs opened by others in the last 3 d in repos the user reviewed (90 d)	1
Received interactions	user ← PR / Issue.author ← IssueComments, PRReviews, PRReviewComments	Replies received on the user's own PRs / issues in 3 and 7 d, and distinct partners (self excluded)	12
Collaborators	user ↔ co-actor on the same PR / issue (cohort only)	Collaborators' 7-day activity per type; number of collaborators (90 d)	7
Static baseline	user → Repositories ← all users (all history)	Degree, concentration, neighbours' activity, all-time replies received	18

Design choices short windows (3 / 7 d) capture current collaboration; 'others' always excludes the user; collaborator activity uses cohort users only, because non-cohort users' activity is incomplete by construction. Submitted model: 7 user + 36 temporal + 32 dynamic relational = 75 inputs.

(3%) Question 2 What the relational signal looks like

The tables keep only cohort users' own events, so replies from others are rarely observed

Why relations add little here: the cohort-only data rarely contains other people's replies



40 / 10,524
comments on cohort-authored threads written by someone else

0.08%
of users have any non-zero received-interaction feature

3–5%
of the submitted model's gain comes from relational features

(3%) Question 2 Relational ablation

Relations do not raise the development Score on top of temporal features

Variant (development weeks 10/22 + 10/29)	Dev Score	Δ vs temporal (95% CI)
Temporal only (7/14/30 d)	35.73%	—
+ static relations	35.02%	−0.71 pt (−1.36, −0.05)
+ received interactions	35.73%	0.00 (features all zero)
+ dynamic relations (submitted)	**35.12%**	**−0.60 pt (−1.17, −0.06)**
dynamic relations − collaborators	35.30%	−0.43 pt (−0.97, +0.15)
Static relations only (no temporal)	33.45%	−2.27 pt (−3.41, −1.13)

Paired user-level bootstrap, 1,000 resamples within each week (10,334 user-weeks). On the TA reference week 09/27–10/04 (higher-activity regime), the submitted model is 53.95% vs 53.79% for temporal only.

Why the relational model is submitted

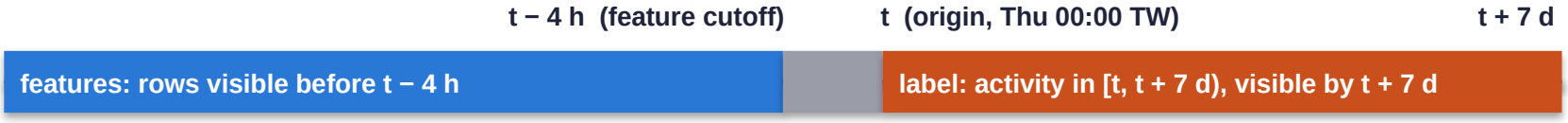
- It was fixed in the config before any real-data run, so the choice is not tuned on development weeks.
- Its gap to temporal-only is < 1 pt, and it stays above copy-last-week (+0.30 pt).

What would make relations useful

- Observing non-cohort replies, or learning over the graph end-to-end: a heterogeneous temporal GNN / RDL (TA GNN with time encoding: 54.82% on the reference week).

(5%) Question 3 As-of sampling prevents leakage

Every training row is rebuilt exactly as the archive looked at prediction time t



Row time, not creation time

A row enters only after its row time (first_seen_at / created_at); PRs, issues and reviews can appear weeks after created_at.

Visible endpoints

The event must occur before the cutoff and its user and repository rows must already be visible.

No later-known columns

merged_at, merged_by_id, closed_at, state_reason and dismissed_at are never used.

Time-ordered splits

Weekly origins; train 09/10–09/17, calibration 10/01–10/08, development 10/22–10/29; each split's labels end before the next split's cutoff.

(5%) Question 3 Temporal features

Multi-window counts, recency and trend for each activity; recency carries most of the signal

Group	Definition (per user, at cutoff t − 4 h)	#
Window counts	log(1 + count) of each activity in the last 7 / 14 / 30 d	18
Activity span	active days and distinct repositories in 7 / 14 / 30 d	6
Recency	days since the last occurrence of each activity (cap 400)	6
Trend	$(count_7 + 1) / (count_{30} / 4 + 1)$ for each activity	6
User profile	days since first seen; log lifetime count of each activity	7

Counts are log-scaled; windows stop at the cutoff, so a 30-day window at an early origin only sees collected days (warm-up limits are logged in coverage.csv).

62–64%

of LightGBM gain from recency for pr_review and pr_review_comment

56%

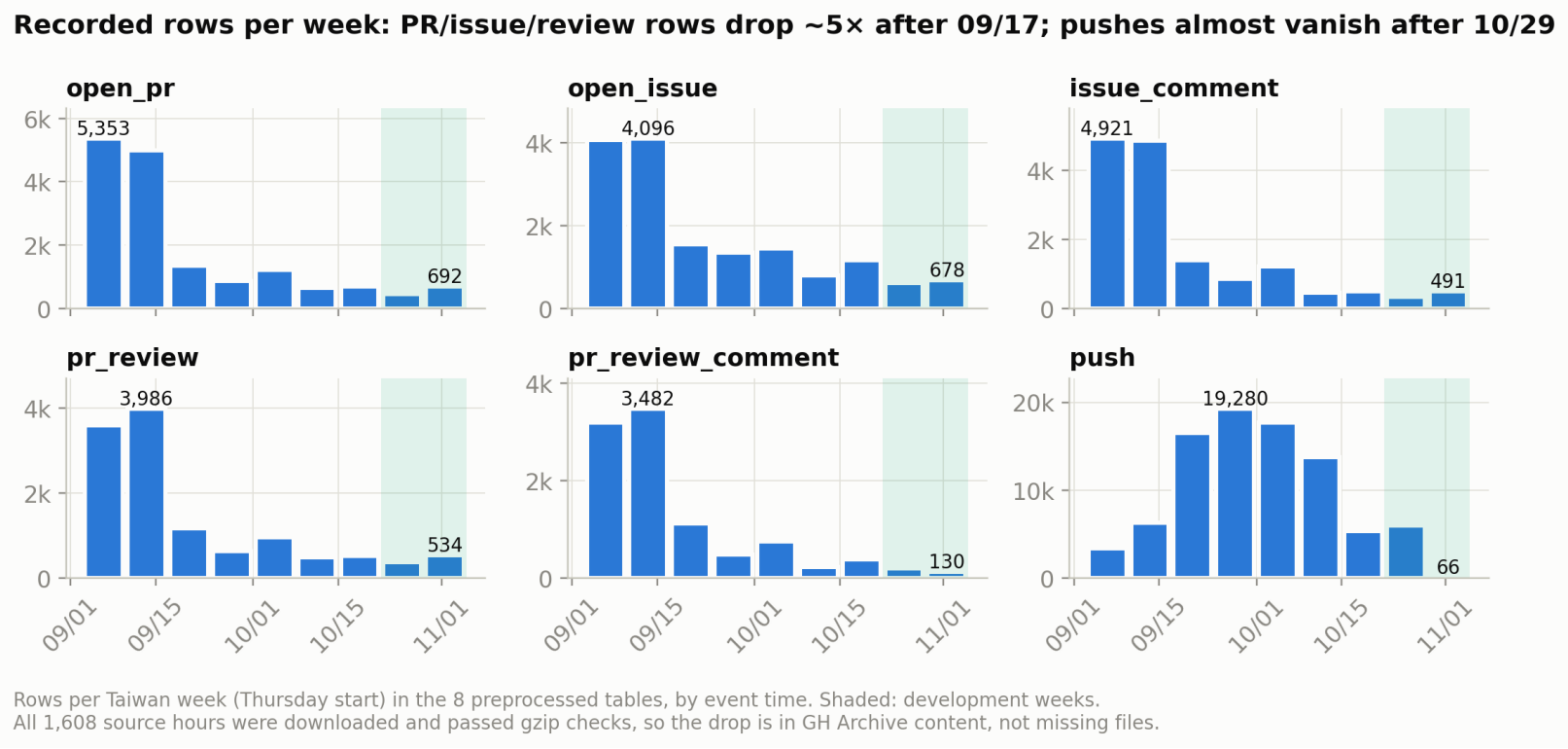
of gain from window counts for push (45% for open_pr)

Top 5 inputs

recency of pr_review, account age, recency of pr_review_comment, open_issue, issue_comment

(5%) Question 3 The data are not stationary

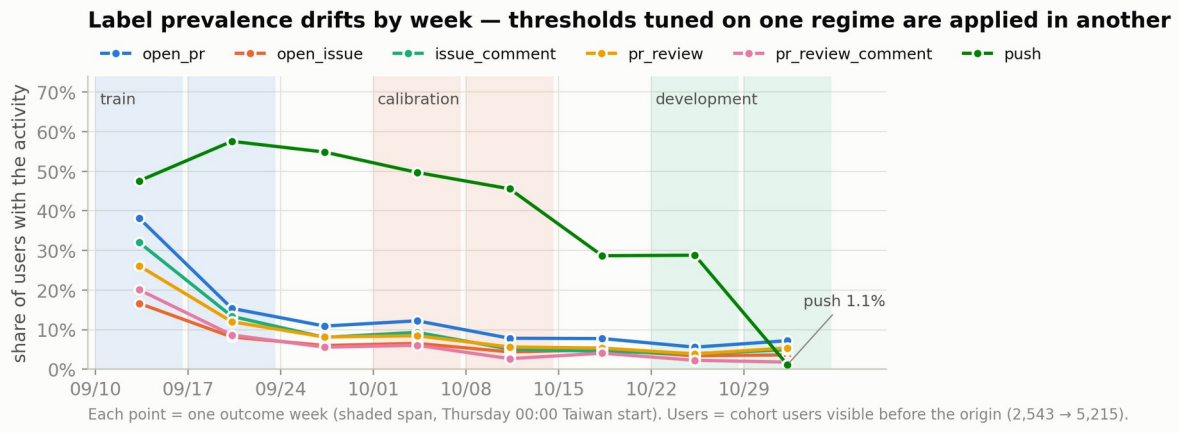
Recorded rows fall ~5× after 09/17 and pushes nearly vanish after 10/29



Rows per Taiwan week (Thursday start) in the 8 preprocessed tables, by event time. Shaded: development weeks.
All 1,608 source hours were downloaded and passed gzip checks, so the drop is in GH Archive content, not missing files.

(5%) Question 3 Label drift and threshold calibration

Positive rates shift by week, so thresholds are tuned per activity on separate calibration weeks



Frozen threshold	PR	Issue	Comment	Review	Line cmt	Push
value	.300	.275	.375	.175	.225	.425

Each threshold maximises the mean weekly Score over the two calibration weeks (coordinate search, 3 passes), then stays fixed for development and the final prediction.

Regime shift

- Calibration Score ≈ 52%, development ≈ 35%.
- open_pr prevalence 38.1% → 5.6% (09/10 → 10/22).
- Push 28.8% → 1.1% in the last week: the six-activity Score falls from 41–45% to 21–24% for every variant; without push it stays at 31–36%.
- Development weeks are never used for tuning, so this gap is reported, not hidden.

(5%) Question 3 Visibility time: the availability clock

Issues often enter the archive long after they were opened; counting by visibility time helps

Table	Rows	Visible > 24 h after event time
Issues (open_issue)	19,379	30.4%
PullRequests (open_pr)	16,694	2.8%
PRReviews (pr_review)	12,524	0.4%
IssueComments, PRReviewComments, Pushes	114,025	0%

Occurrence clock

windows ask “what happened in the last k days?” — an old issue that only just became visible is counted in its original week.

Availability clock

windows ask “what newly became visible in the last k days?” — the view the archive actually offers at prediction time.

36.41%

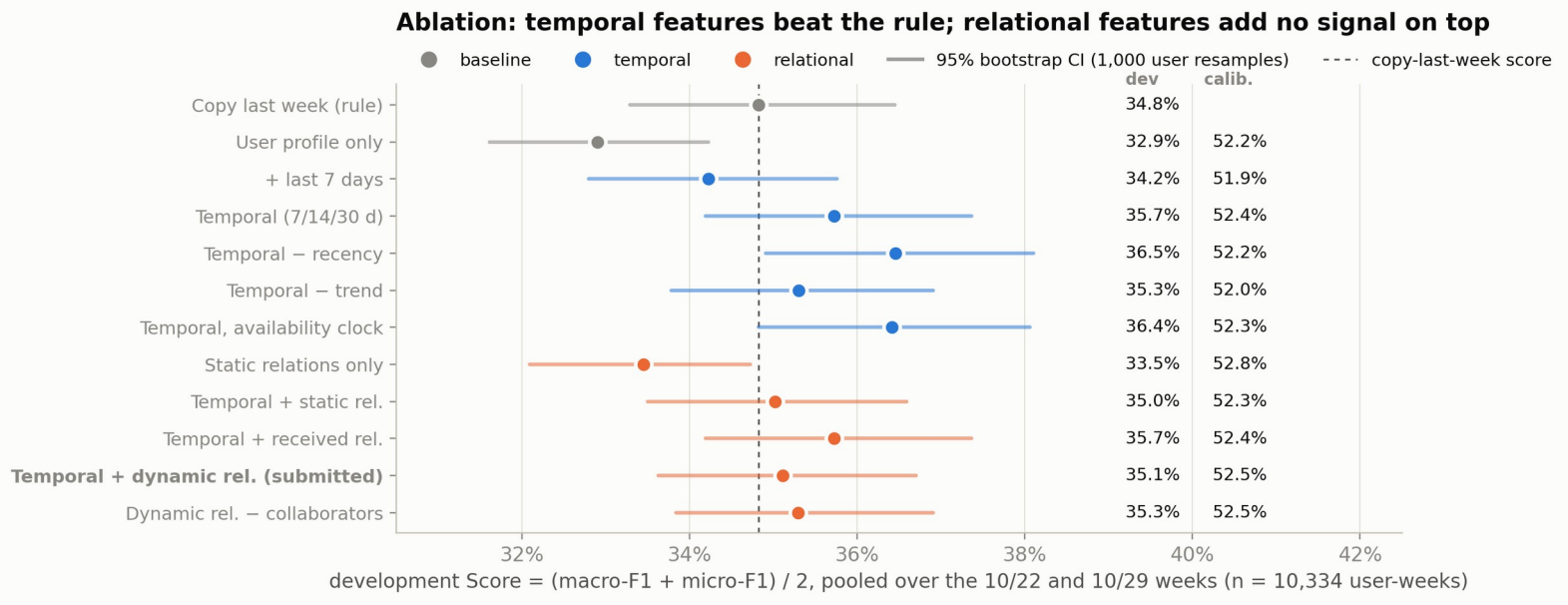
development Score with the availability clock (best temporal variants: 36.4–36.5%)

+1.59 pt

vs copy-last-week;
95% CI +0.62 to +2.62 pt

(5%) Question 3 Ablation with bootstrap CIs

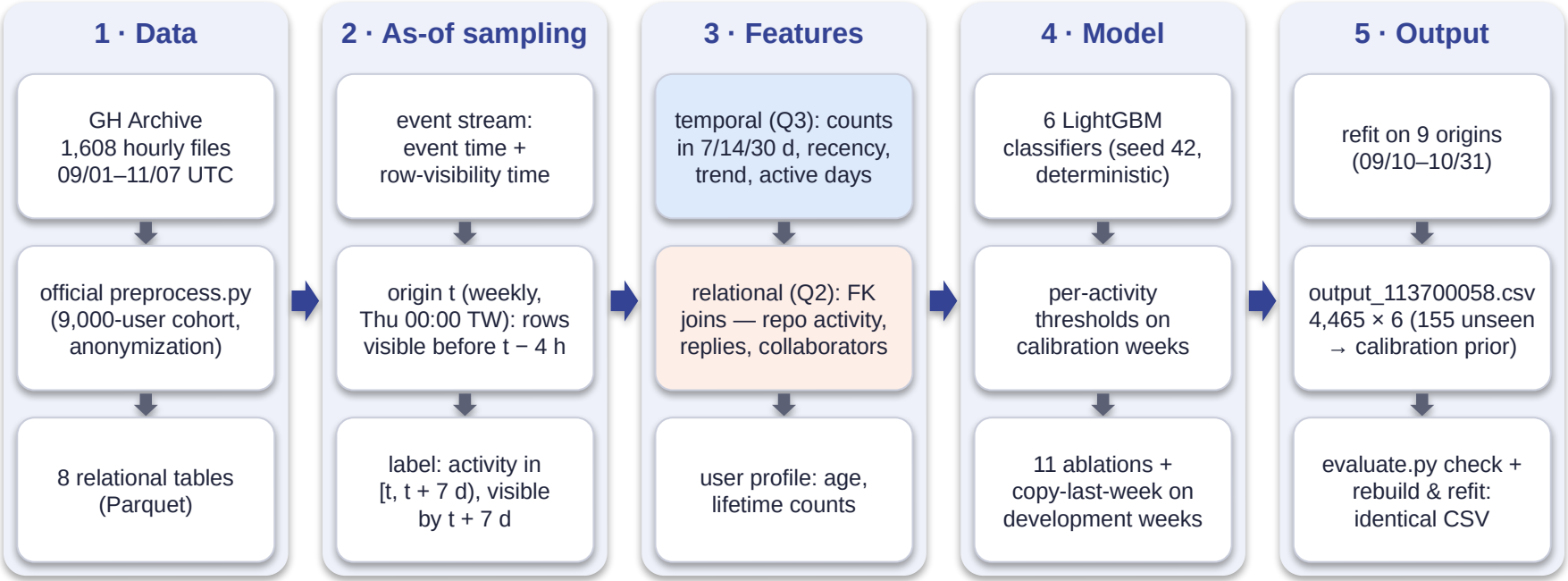
Temporal information is what beats copy-last-week; all variants share dates, users, labels, tuning



Δ vs copy-last-week: user profile only −1.89 pt (CI excludes 0) · + 7/14/30-day windows, recency, trend +0.91 pt (P(better) = 0.97) · removing recency +1.64 pt and availability clock +1.59 pt (both CIs exclude 0). Dev. weeks 10/22 + 10/29, 10,334 user-weeks, 1,000 paired user resamples.

(5%) Question 4 Framework

From GH Archive hours to a checked, reproducible submission file



Blue = temporal (Q3), orange = relational (Q2). The same as-of sampler produces training, calibration, development and test rows, so prediction-time features are computed exactly as in training.

(5%) Question 4 Pipeline steps and evaluation

Six steps, each with a check; our labels reproduce the TA's copy-last-week reference

1. **Download** 1,608 hourly files; SHA-256 recorded; gzip verified; resumable.
2. **Build tables** unchanged official preprocess.py → 8 tables; PK/FK checks.
3. **Sample** weekly origins (Thu 00:00 TW); as-of features + 7-day labels.
4. **Train** 6 LightGBM heads: 160 trees, lr 0.045, 23 leaves, seed 42, deterministic.
5. **Calibrate + evaluate** thresholds on 10/01 + 10/08; 11 ablations on 10/22 + 10/29.
6. **Refit + predict** 9 origins 09/10–10/31; 4,465 users; 155 unseen → calibration prior.

Reference week 09/27–10/04	TA README	Ours
Copy last week	49.61%	50.39%
GNN with time encoding	54.82%	—
Temporal (7/14/30 d)	—	53.79%
Temporal + dynamic rel. (submitted)	—	53.95%

- Our copy-last-week score is within 0.8 pt of the README value, so our label construction is consistent with the TA's.
- Our model rows are optimistic: this week overlaps the calibration window.

(5%) Question 4 Output and conclusions

The submitted CSV can be regenerated byte for byte from GH Archive with the code on GitHub

4,465 × 6

rows in input order; official evaluate.py format check passes

Identical CSV

after rebuilding all 8 tables and refitting (SHA-256 f5be6efe…)

Identical model

checkpoint re-trained on a second machine

Conclusions

- Time is the main signal: multi-window counts and recency beat copy-last-week; visibility-aware counting helps most under drift.
- Relational aggregation adds little here because cohort-only tables rarely contain other users' replies (40 of 10,524 comments).
- Strong distribution shift (volume −5×, push → 1%) dominates the scores; per-activity thresholds on separate weeks limit the damage.

Limitations and next steps

- Two development weeks: most differences are within ±1.5 pt and CIs overlap.
- Next: a heterogeneous temporal GNN / RDL model and drift-aware threshold updates.

Code and results: github.com/ellatso/NYCU-BigData-Assignment2